

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0763

Mcindoes 138 kV
August 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0763. The Interconnection Customer proposes a 74.4 MW generating facility interconnecting at the Mcindoes 138 kV substation in the Virginia area. The request was received on March 24, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 302 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$250,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Redbed Plains Wind Farm – Westcoast Cellufibre 138.0 kV
Contingency: Mitchell Dam – Skaggs Island Jct 2 345.0 kV Line
Loading: 143.5% (Rating: 230 MVA)
DFAX: 5.9%
Pre-project loading: 141.6%
2. Facility: McCartney – Roanoke Rapids 345.0 kV
Contingency: Roanoke Rapids – Gaston 345.0 kV
Loading: 115.2% (Rating: 1,398 MVA)
DFAX: 7.1%
Pre-project loading: 114.9%
3. Facility: Walthall – Haggin 138.0 kV
Contingency: North Tower Tap – Haggin 345.0 kV
Loading: 108.4% (Rating: 439 MVA)
DFAX: 6.4%
Pre-project loading: 107.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Redbed Plains Wind Farm – Westcoast Cellufibre 138.0 kV	Rebuild line with higher-capacity conductor	\$4,110,000	2% (\$71,000)
Roanoke Rapids Substation – McCartney 345.0 kV line	Rebuild line with higher-capacity conductor	\$49,205,000	0% (\$163,000)
Haggin – Walthall 138.0 kV	Reconductor line	\$1,569,000	1% (\$16,000)
	Total allocated to project		\$250,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.